

Activity 3: Cluster Interpretation Document

Student Segmentation & Risk Detection using Machine Learning

1. Objective

The objective of this activity is to apply unsupervised machine learning techniques to identify hidden patterns in student performance, behavior, and well-being data. By using K-Means clustering, students are grouped into meaningful segments to detect risk levels and support personalized interventions. The results are also compared with the rule-based Student Risk Index (SRI).

2. Dataset Description

The dataset used in this activity is the scored student dataset obtained from Activity 2. It includes academic, behavioral, engagement, and well-being indicators.

Key attributes used for clustering:

- Attendance percentage
- Assignment completion rate
- Engagement score
- Academic Performance Score (APS)
- Sleep score
- Stress score

The dataset also contains a rule-based SRI category (Red, Yellow, Green), which is used only for comparison and not for clustering.

3. Methodology

- Algorithm Used: K-Means Clustering
- Type: Unsupervised Machine Learning
- Data Preprocessing: Feature normalization using StandardScaler
- Optimal Number of Clusters: Determined using the Elbow Method (K = 3)
- Visualization Technique: Principal Component Analysis (PCA)

K-Means was chosen due to its effectiveness in grouping students based on similarity in behavioral and performance attributes.

4. Cluster Interpretation

Cluster 0 – At-Risk Students

Characteristics:

- Low attendance and assignment completion
- Low engagement levels
- High stress scores

- Poor lifestyle balance (sleep and well-being)

Interpretation:

Students in this cluster show signs of academic struggle and mental pressure. These students are at high risk of poor performance, burnout, or dropout and require immediate academic and emotional support.

Cluster 1 – High Performers

Characteristics:

- High attendance and engagement
- Strong academic performance (high APS)
- Balanced sleep and stress levels
- Positive behavioral patterns

Interpretation:

This cluster represents consistently performing students who are academically strong and mentally stable. They are well-prepared for advanced learning opportunities and career development programs.

Cluster 2 – Career-Confused Students

Characteristics:

- Moderate academic performance
- Low engagement despite adequate attendance
- Average stress levels
- Inconsistent behavioral indicators

Interpretation:

Students in this cluster have the potential to perform well but lack clarity in career goals or motivation. Without guidance, these students may underperform in the long term.

5. Comparison: Machine Learning vs Rule-Based SRI

- Rule-based SRI relies on fixed thresholds and predefined conditions.
- Machine learning clustering identifies hidden patterns across multiple dimensions simultaneously.
- Some students classified as “Yellow” in SRI are identified as “At-Risk” by ML due to behavioral and stress indicators.
- ML provides early detection of risk that may not be visible through rule-based scoring.

6. Conclusion

The machine learning-based clustering approach provides deeper and more dynamic insights into student behavior and risk levels compared to traditional rule-based systems. By identifying At-Risk, High Performer, and Career-Confused students, institutions can design targeted interventions that improve academic outcomes and student well-being.